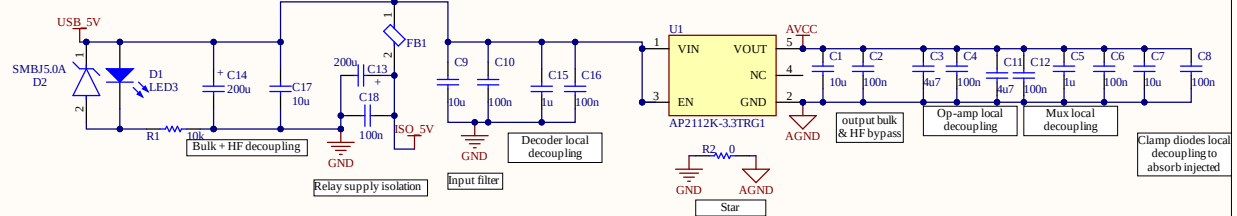


32-RANGE OSCILLOSCOPE ANALOG FRONT END

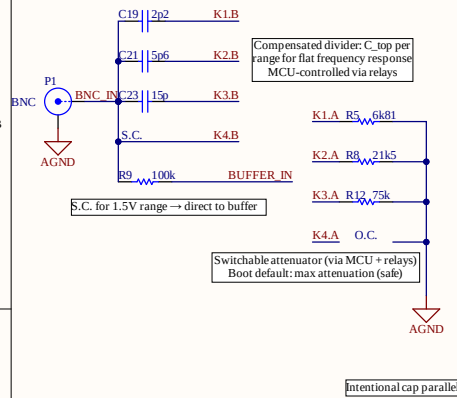
Single-channel AFE for STM32F411 12-bit ADC (0–3.3V) | BW ~500kHz | Input: ±24V → ±275mV

Auto-ranging: MCU starts at max attenuation, steps to higher gain until near full-scale

POWER SUPPLY AND DISTRIBUTION

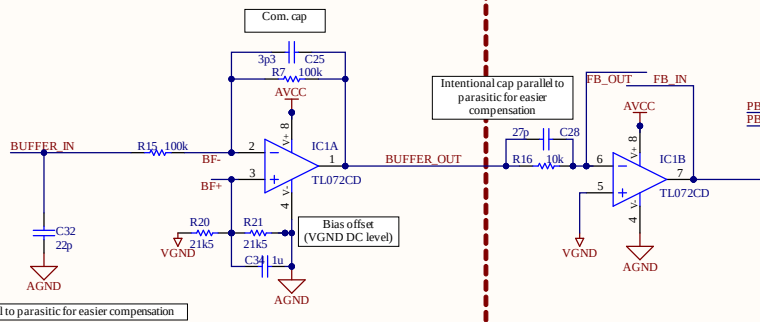


SWITCHABLE ATTENUATOR

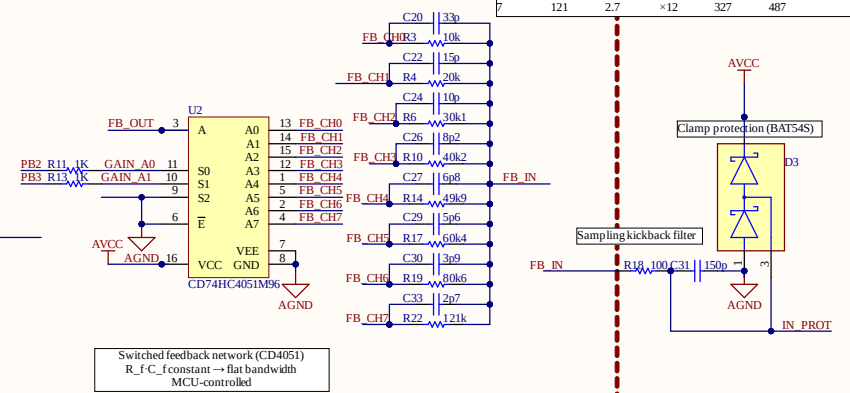


INVERTING BUFFER/LEVEL SHIFTER

Gain = -1, Level shift to V_{GND}

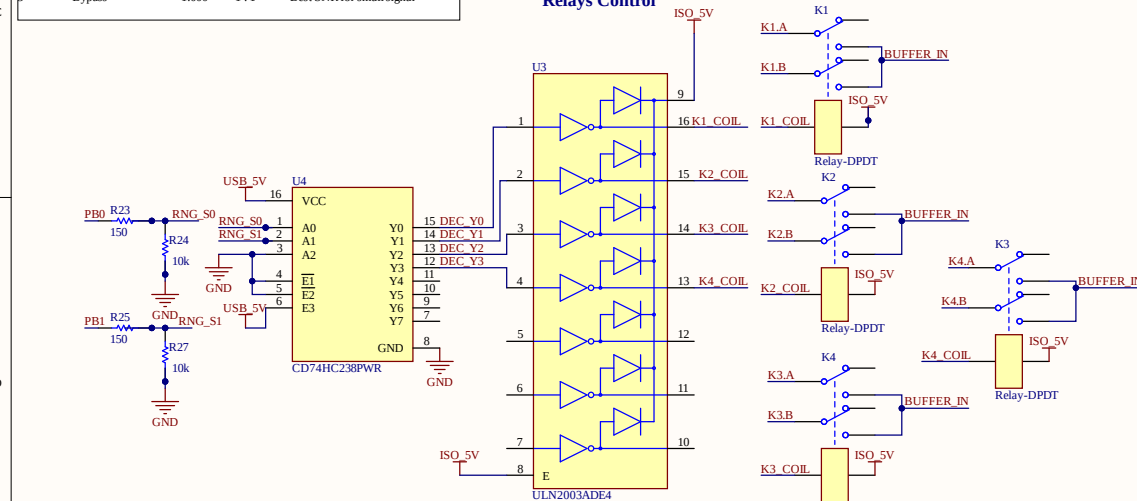


PGA (MCU-controlled) Can use integrated alternative (e.g. AD8253)

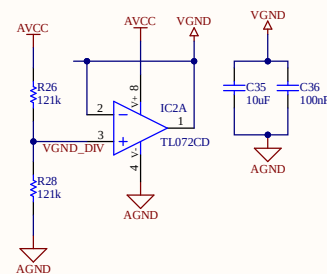


Ch	Rf (kΩ)	Cf (pF)	Gain	Rf × Cf (ns)	f _{-3dB} (kHz)
0	10.0	33	×1	330	482
1	20.0	15	×2	300	531
2	30.1	10	×3	301	529
3	40.2	8.2	×4	330	483
4	49.9	6.8	×5	339	469
5	60.4	5.6	×6	338	471
6	80.6	3.9	×8	314	507
7	121	2.7	×12	327	487

Relays Control



VGND Generator (1.65V mid-rail)



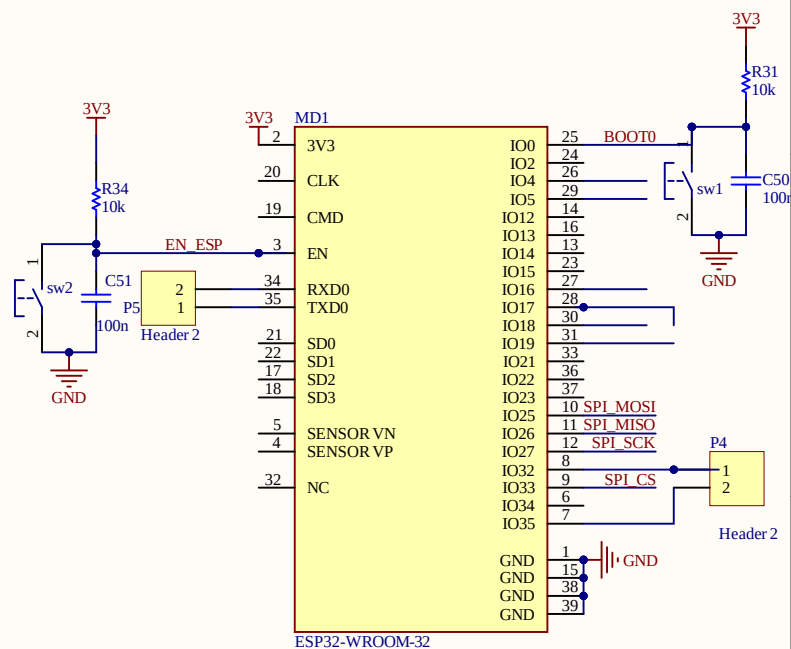
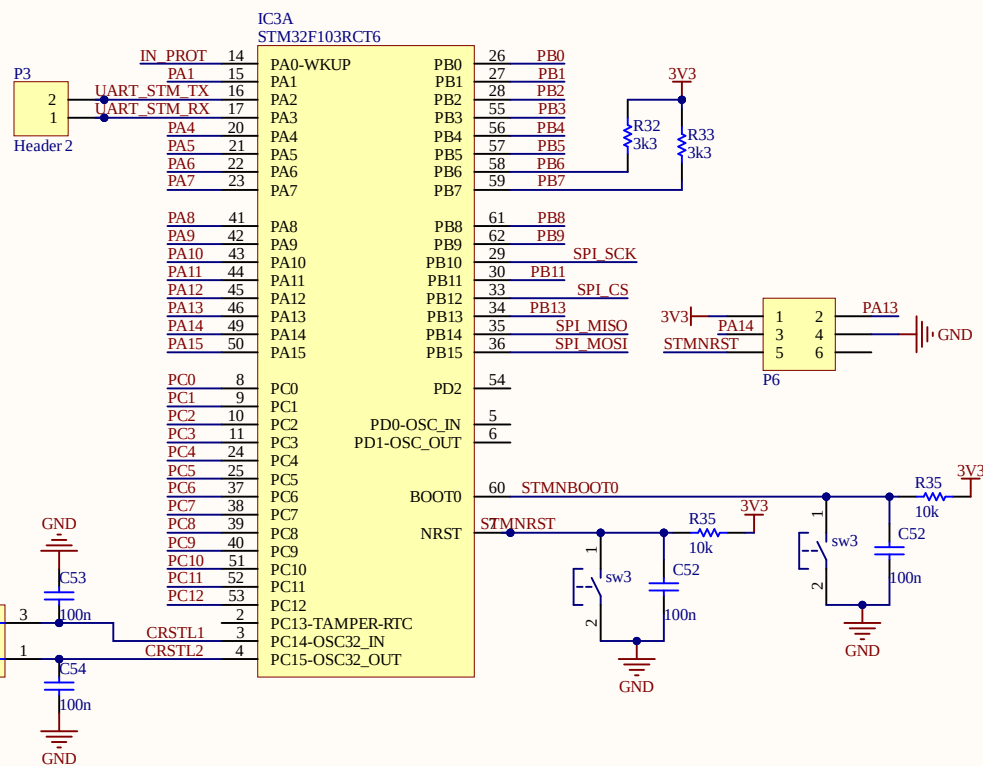
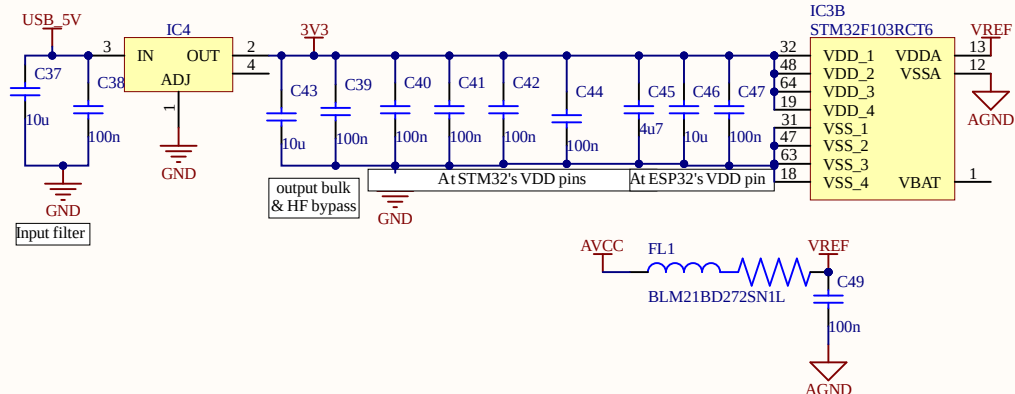
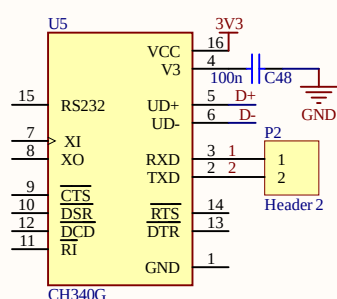
MCU AUTO-RANGE ALGORITHM:

Start at max attenuation → step up until ADC near full-scale

Input Amplitude (Vpp)	Preferred Divider	Preferred Gain	Effective FS
< 275 mV	Bypass	×12	275 mV
275 mV – 413 mV	Bypass	×8	413 mV
413 mV – 550 mV	Bypass	×6	550 mV
550 mV – 660 mV	Bypass	×5	660 mV
660 mV – 825 mV	Bypass	×4	825 mV
825 mV – 1.10 V	Bypass	×3	1.10 V
1.10 V – 1.65 V	Bypass	×2	1.65 V
1.65 V – 3.30 V	Bypass	×1	3.30 V
3.30 V – 7.70 V	75k	×1	7.70 V
7.70 V – 18.6 V	21.5k	×1	18.6 V
18.6 V – 51.7 V (BOOT)	6.81k	×1	51.7 V

Title		
Size	Number	Revision
A3		
Date:	2/13/2026	Sheet of
File:	C:\Users\...\AFE_SchDoc	Drawn By:

USB-C / CC Pull-downs (5.1kΩ)



Title		
Size A4	Number	Revision
Date:	2/13/2026	Sheet of
File:	C:\Users\...\Digital\SchDoc	Drawn By: